

Why do protists make poor mathematicians? Each question is worth 2 points except where noted. Good luck!

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

1. The type of nerve cell responsible for integrating different stimuli is the
 - A. sensory neuron
 - B. axon
 - C. interneuron**
 - D. motor neuron
 - E. dendrite
 2. Transpiration in plants requires all of the following *except*
 - A. evaporation of water molecules.
 - B. active transport through xylem cells.**
 - C. cohesion between water molecules.
 - D. adhesion of water molecules to cellulose.
 - E. water vapor on the spongy mesophyll.
 3. Ignoring all other factors, what kind of day would result in the fastest delivery of water and minerals to the leaves of a tree?
 - A. A cool, dry day.
 - B. A very hot, dry, windy day.
 - C. A warm, dry day.**
 - D. A warm, humid day.
 - E. A cool, humid day.
 4. Which of the following are primarily responsible for movement in the eukaryotic flagellum?
 - A. Basal apparatus and hook.
 - B. Dynein and microtubules.**
 - C. Actin and myosin filaments.
 - D. Hydrogen ions and basal apparatus.
 - E. Sensory and motor neurons.
 5. Hydrostatic skeletons are
 - A. external skeletons that provide support and protection against dehydration.
 - B. fluid-filled compartments under pressure.**
 - C. skeletons found in deep sea organisms that depend on high pressure of the surrounding water to provide support.
 - D. internal skeletons that are buried in soft tissue.
 - E. skeletons made of microtubules in the cytoplasm of protists.
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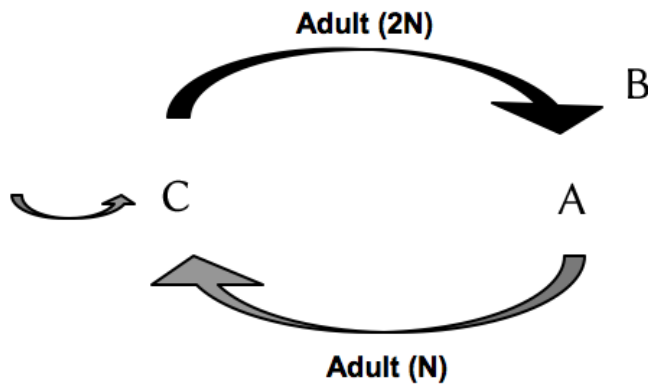
6. Galvanotaxis occurs when
- A. protists respond to touch.
 - B. animals respond to temperature.
 - C. protists respond to an electrical field.**
 - D. plants respond to gravity.
 - E. plants respond to hormones.
 - F. animals respond to metal ions.
7. Pores on the leaf surface that function in gas and water exchange are called
- A. stomata.**
 - B. waxy cuticles.
 - C. spongy mesophyll.
 - D. phloem.
 - E. xylem.
8. In terms of structure, cilia are most like
- A. microtubules.
 - B. myosin filaments.
 - C. actin filaments.
 - D. bacterial flagella.
 - E. eukaryotic flagella.**
9. The ion used most often as a second messenger is
- A. K^+
 - B. H^+
 - C. Na^+
 - D. Cl^-
 - E. Ca^{+2}**
10. The fluid that moves around in the circulatory system of a typical arthropod is the
- A. cytoplasm.
 - B. blood plasma.
 - C. hemolymph.**
 - D. intracellular fluid.
 - E. gastrovascular fluid.
11. Most of the water taken up by a plant
- A. transports sugars in the phloem.
 - B. keep guard cells turgid.
 - C. is lost during transpiration.**
 - D. is converted to sugars during photosynthesis.
 - E. enters the central vacuoles of cells.
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12. Organisms with a circulating fluid that is separate from the fluid that directly surrounding the body's cells are likely to have
- A. a hemolymph system.
 - B. a vascular system.
 - C. a closed circulatory system.**
 - D. an open circulatory system.
 - E. a gastrovascular cavity.
13. The main stimulus used by birds to find their way north for the summer time is
- A. electromagnetic.**
 - B. thigmic.
 - C. chemical.
 - D. thermal.
 - E. mechanical.
14. What drives the flow of water through the xylem?
- A. The number of companion cells in the phloem.
 - B. Abscissic acid opening guard cells around the stomata.
 - C. Transpirational pull through sieve-tube elements.
 - D. The evaporation of water from the leaves.**
 - E. A higher concentration of sugars in the leaves than in the roots.
15. During protist conjugation, which of the following does *not* occur during some part of the process?
- A. Some micronuclei disintegrate.
 - B. The macronucleus fuses with the micronucleus.**
 - C. The micronucleus undergoes mitosis.
 - D. The macronucleus disintegrates.
 - E. Cells exchange micronuclei.
 - F. The micronucleus undergoes meiosis.
16. From an energy perspective, animals are mobile because they consume
- A. low quality energy and low energy efficiency.
 - B. high quality energy and carbon intake is irregular.
 - C. low quality energy and carbon intake is constant.
 - D. low quality energy and carbon intake is irregular.**
 - E. high quality energy and carbon intake is constant.
 - F. high quality energy and high energy efficiency.
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17. The highest quality form of energy (best able to do work) for organisms is
- A. thermal.
 - B. radiant.**
 - C. chemical.
 - D. potential.
 - E. kinetic.
18. When ATP attaches to the myosin head
- A. the actin head attaches to the myosin filament.
 - B. the myosin head releases from the actin filament.**
 - C. the myosin head initiates the power stroke.
 - D. the actin head attaches to the myosin filament.
 - E. the myosin head attaches to the actin filament.
19. The protein that is activated in response to auxin that allows phototropism to occur is
- A. transmembrane.
 - B. wallulase.
 - C. basal apparatus.
 - D. abscisic acid.
 - E. expansin.**
 - F. proton channel.
20. The vascular tissue system for delivery of water, minerals, and nutrients throughout the plant includes all of the following except for
- A. phloem.
 - B. xylem.
 - C. cohesion.**
 - D. companion cells.
 - E. sieve plates.
21. Which of the following about phloem is false?
- A. Sugars move by passive diffusion.
 - B. Companion cells provide proteins to sieve-tube elements.
 - C. The cell has cytoplasm.
 - D. Sieve-tube elements do not have a nucleus.
 - E. The plasma membrane is modified to form a sieve plate.**
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22. Differences in energy quality and availability affects the amount of surface area and volume in organisms. Compared to animals, plants have
- A. low surface area and high volumes.
 - B. high surface area and low volumes.**
 - C. high surface area and high volumes.
 - D. low surface areas and low volumes.
23. The function of valves in veins in a closed circulatory system is to
- A. enhance diffusion of nutrients to the tissues.
 - B. ensure one way flow of circulatory fluids.**
 - C. enable the vein to withstand high pressure.
 - D. provide support for the extra muscles around the vein.
 - E. decrease the amount of pressure in the veins.

Use the following figure to fill in the blanks of the next four questions.



24. The stage at A (that results from B) is called **Spore**
25. The process at B is **Meiosis**
26. The process (not the stage) that occurs at C is **Syngamy**
27. Which of the three (A, B or C) does not exist in the animal lifecycle. **A**

Keep on keepin' on!

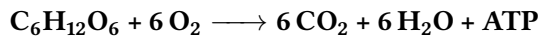
Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Active Diffusion	Gametophyte	Pseudopodia
Amplification	Gastrovascular Cavity	Reception
Binary Fission	Haplogamy	Response
Chemotaxis	Karyogamy	Run and Tumble
Cytokinesis	Magnification	Second Messenger
Cytoplasmic Streaming	Mechanotaxis	Signalation
Cytotaxis	Meiosis	Sporophyte
Diplogamy	Mitosis	Syngamy
Dive and Bomb	Netflix and Chill	Thigmotaxis
Drive and Orient	Passive Diffusion	Touchyfeelytaxis
Galvanotaxis	Plasmosyngamy	Transduction

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|----------------------------------|--|
| 28. Second Messenger | A receptor activates these ions inside a cell. |
| 29. Gametophyte | Mosses spend most of their life in this life cycle stage. |
| 30. Gastrovascular Cavity | Responsible for digestion and nutrient distribution in flatworms. |
| 31. Plasmosyngamy | Life cycle stage where fungal cells have fused but nuclei have not. |
| 32. Thigmotaxis | Locomotion in response to touch. |
| 33. Amplification | Increases the strength of a weak signal. |
| 34. Transduction | A signal is transmitted to a target area inside a cell. |
| 35. Cytokinesis | Division of the cell after mitosis. |
| 36. Cytoplasmic Streaming | Transports nutrients throughout a cell. |
| 37. Run and Tumble | Allows bacteria to move towards a chemical cue. |
| 38. ANY ANSWER | Name the band that you think is Taylor's favorite (not in list above). |

Circle True or False. Read each statement completely before answering.

39. **TRUE** False Without H^+ ions bacteria could not move their flagella.
40. True **FALSE** High humidity increases the rate of transpiration.
41. **TRUE** False Adhesion is the hydrogen bonding of water molecules to the cell walls of the xylem.
42. True **FALSE** A signal is not received at the organismal level in protists or bacteria.
43. True **FALSE** Fungi and animals both spend most of their life cycle in the diploid stage.
44. **TRUE** False Dynein is the motor protein found in eukaryotic flagella.
45. True **FALSE** Plants spend most of their energy needs for photosynthesis.
46. True **FALSE** Plants seedlings display negative gravitaxis by growing away from gravity.
47. **TRUE** False Open circulatory systems are less efficient than closed circulatory systems.
48. True **FALSE** A chemoheterotrophic protist is more likely to have an eyespot than a photoautotrophic protist.
49. (4 points) You knew it was coming...Write a *balanced* chemical equation for cellular respiration. In addition, include the molecule that is used to accomplish work in the cell.



Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H_2O , 2 pts if more than 1 is wrong.

Subtract 1 pt if one chemical formula is wrong (e.g., $C_{12}H_6O_6$), 2 pts if one missing or more than 1 wrong. Subtract 2 pt if missing ATP.

Ask me if uncertain.

Because they have to divide to multiply. Be wise, do good things, have a great summer!
